

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410UA0-1



Z22-3410UA0-1

FRIDAY, 17 JUNE 2022 – AFTERNOON

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources

HIGHER TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 9(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

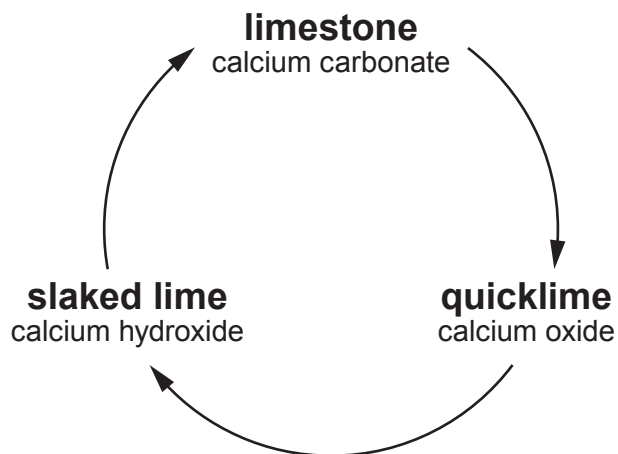
For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	9	
3.	6	
4.	9	
5.	5	
6.	8	
7.	5	
8.	7	
9.	11	
10.	9	
11.	6	
Total	80	

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Answer **all** questions.

1. Limestone is a rock which consists mostly of calcium carbonate. The diagram shows a cycle of reactions involving limestone.



- (a) (i) When limestone is heated, calcium carbonate is converted to calcium oxide and carbon dioxide.

I. State the name for this type of reaction. [1]

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II. Write a balanced symbol equation for the reaction. [2]

..... → +

(ii) State what must be added to calcium oxide to form calcium hydroxide. [1]

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(b) Give **one** use of limestone in the construction industry. [1]

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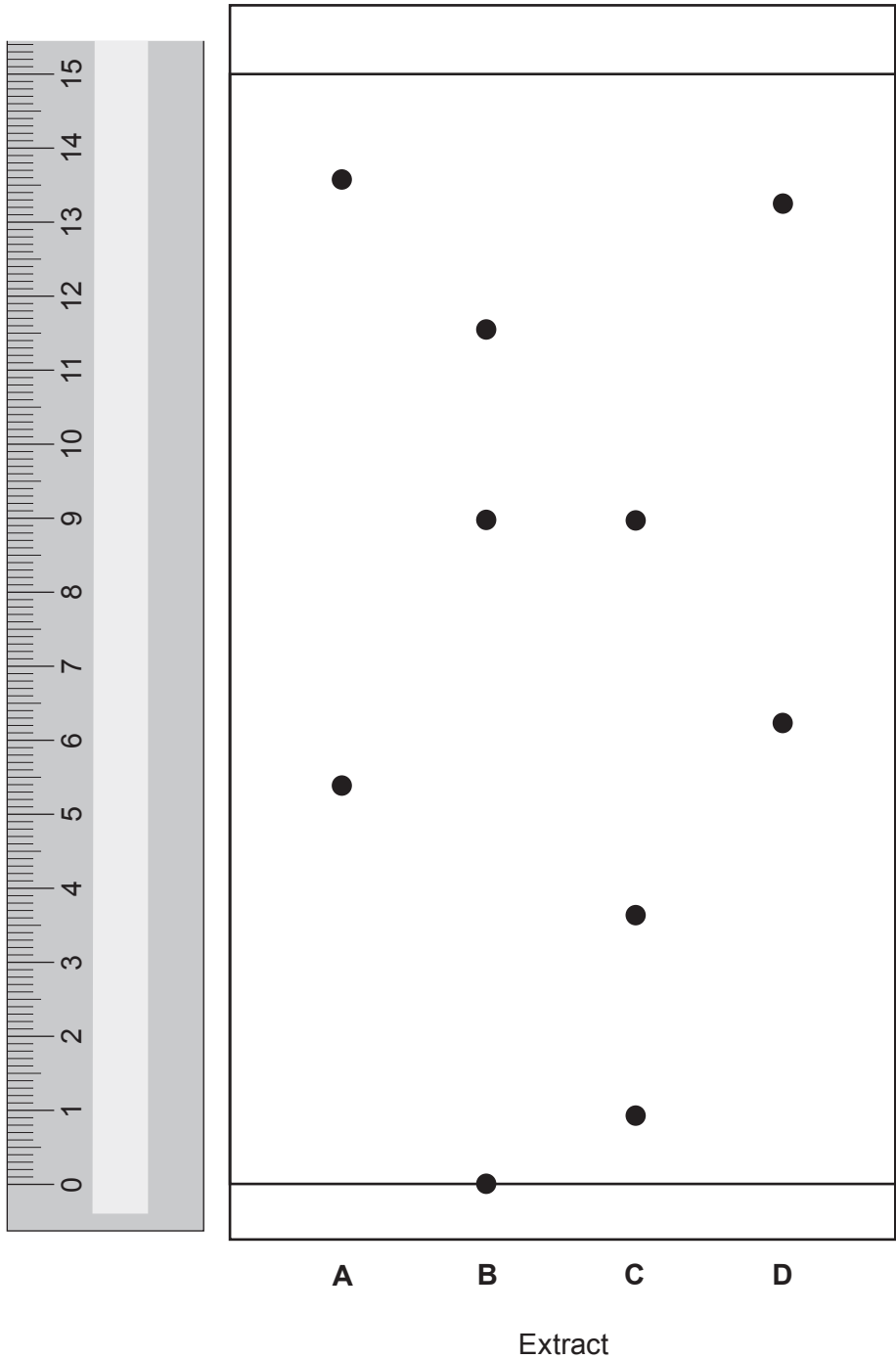
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2. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



(a) All of the extracts contain a mixture of pigments with different R_f values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest** R_f value 0.60?

Give your reasoning. [3]

Extract

Reasoning

.....
.....

(b) Explain why the pigments travel different distances on the chromatogram. [2]

.....
.....
.....

(c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice. [2]

Extract

Explanation

.....

(d) The chemical formula of the solvent acetone is C_3H_6O . Calculate the percentage by mass of carbon in acetone.

The relative formula mass (M_r) of acetone is 58. [2]

$A_r(C) = 12$

Percentage = %

9

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Examiner
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3. (a) The Earth’s early atmosphere contained large amounts of water vapour and carbon dioxide. Explain why the amounts of water vapour and carbon dioxide decreased over geological time. [4]

Water vapour

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Carbon dioxide

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- (b) State the percentages of nitrogen and oxygen in the present atmosphere. [2]

Nitrogen %

Oxygen %

6



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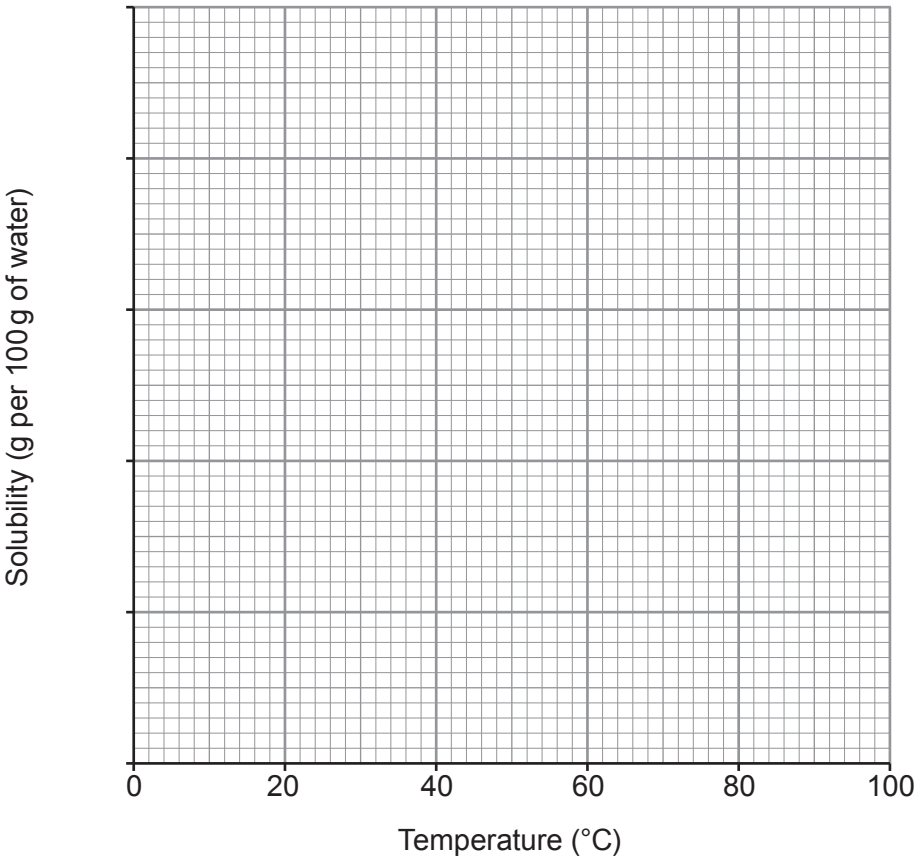
4. (a) The table shows the solubility of potassium nitrate in water at different temperatures.

Temperature (°C)	Solubility (g per 100 g of water)
0	13
20	32
40	64
60	110
80	169
100	246

(i) Choose a suitable scale for the y-axis and plot the data on the grid.

Draw a line of best fit.

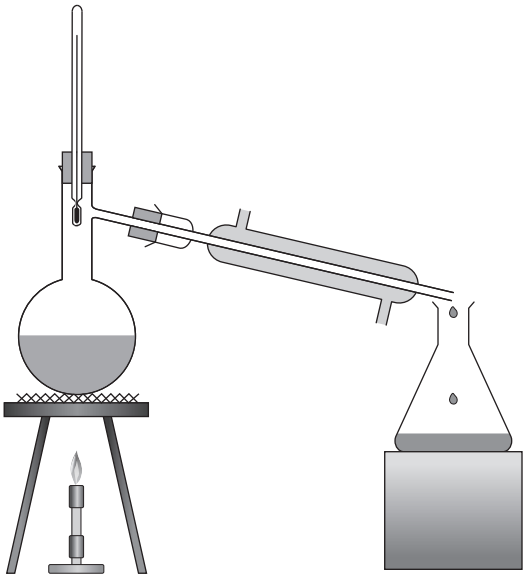
[4]



(ii) Calculate the mass in grams of potassium nitrate that would be crystallised if 250 cm³ of a saturated solution were cooled from 80 °C to 30 °C. [3]

Mass = g

(b) Ethanol can be separated from water by distillation using the apparatus shown.



Explain how ethanol and water are separated by distillation. [2]

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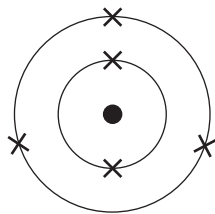
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5. (a) The diagram shows an atom of element **X**.



Draw the electronic structure of the element directly **below** element **X** in the Periodic Table.

[1]



(b) The table below shows information about particles **A-F**.

The letters **A-F** are **not** the chemical symbols of the particles.

Particle	Number of protons	Number of electrons	Number of neutrons
A	9	10	10
B	6	6	8
C	7	7	7
D	6	6	6
E	9	9	10
F	3	2	4

(i) Give the **letters** of **two** particles which are **isotopes** of the same element.
Explain your answer.

[2]

Letters and

.....

.....

(ii) Give the **letters** of **two** particles which are **ions**. Explain your answer.

[2]

Letters and

.....

.....

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5



6. (a) **S** and **T** are two metal halide compounds. The ions in each compound were identified by observations from a flame test and the addition of silver nitrate solution.

(i) Complete the table.

[4]

Compound	Flame test colour	Symbol of ion	Observation on adding silver nitrate solution	Symbol of ion
S	brick red		yellow precipitate	
T		Ba ²⁺	white precipitate	

- (ii) Suggest why a silver nitrate test would not be able to distinguish clearly which halide ions are present in a **mixture** of compounds **S** and **T**.

[1]

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- (b) Write an **ionic** equation for the formation of the precipitate in the reaction between sodium chloride solution and silver nitrate solution.

Include state symbols.

[3]

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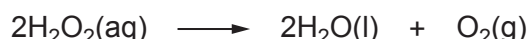
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7. A catalyst is a substance that causes significant changes to the rate of a chemical reaction. Catalysts interact with the reactants in a chemical reaction and eventually release the product allowing the catalyst to be recovered.

Enzymes are biological molecules which catalyse reactions in living organisms. Enzymes tend to have an optimum temperature of 37 °C.

Hydrogen peroxide is a colourless liquid with the chemical formula H_2O_2 . It has many uses including as a bleach and an antiseptic. It is also found in all animals and plants. The decomposition of hydrogen peroxide happens very slowly on its own.



The reaction can be catalysed by several substances including iron(III) oxide, Fe_2O_3 , manganese dioxide, MnO_2 , and the enzyme catalase.

Four catalysts were tested to determine which was the most efficient. An equal mass of each catalyst was added to equal volumes of hydrogen peroxide solution of equal concentration. The volume of oxygen produced in 1 minute was recorded. The experiment was repeated at different temperatures.

Temperature (°C)	Volume of oxygen produced in 1 minute (cm ³)			
	Catalyst W	Catalyst X	Catalyst Y	Catalyst Z
10	0.8	1.2	0.4	0.9
20	1.4	1.4	0.9	1.5
30	1.9	2.1	1.3	2.1
40	2.5	2.6	1.6	2.5
50	3.4	3.8	1.2	3.5
60	4.6	4.8	0.7	4.7
70	5.2	5.4	0.3	5.3



(a) Give the **number** of the statement which is **not** correct. [1]

- 1 Catalyst **W** is more efficient than **Y** but less efficient than **X** and **Z**
- 2 Catalyst **X** is the most efficient of the four
- 3 Catalyst **Y** is less efficient than all of the others
- 4 Catalyst **Z** is more efficient than **Y** but less efficient than **W** and **X**

.....

(b) State which catalyst, **W**, **X**, **Y** or **Z**, could be catalase. Explain your answer. [3]

Catalyst

.....

(c) Three students made statements about the properties of catalysts.

- Katrina said that a catalyst does not get used up in the reaction but makes it happen faster.
- Jo said that the reactants are used up in less time because a catalyst makes a different product.
- Rhys said that adding a catalyst allows the reaction to happen with a lower activation energy, but the same products are made.

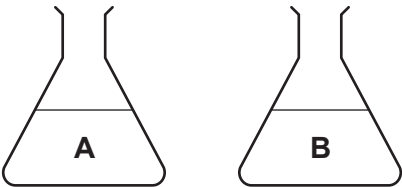
Which of the students' statements are correct? [1]

- A** Jo only
- B** Rhys and Jo, but not Katrina
- C** Katrina and Rhys, but not Jo
- D** All three students

.....



8. Hard water can be softened using different methods, including boiling the water and passing it through an ion exchange column.
- (a) Two hard water samples, **A** and **B**, were tested for hardness, using soap solution. Both samples were tested before and after boiling, and after passing through an ion exchange column.



	Volume of soap solution needed to produce a lather (cm ³)		
Water sample	Before boiling	After boiling	After ion exchange
A	15	15	2
B	17	9	2

State the type (or types) of hard water that each sample contains. Give your reasoning. [4]

A

.....

.....

B

.....

.....

.....

.....



- (b) Sodium carbonate (washing soda) can also be used to soften hard water. It reacts with the magnesium compounds in the water.

Write a balanced symbol equation for the reaction between sodium carbonate and magnesium chloride, to form sodium chloride and magnesium carbonate. [3]

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Examiner
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Examiner
only

9. (a) Give the observations made during the reactions of lithium, sodium and potassium with water. Explain the trend in reactivity in terms of electronic structure. [6 QER]

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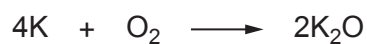
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- (b) Potassium reacts with oxygen to produce potassium oxide.



- (i) Calculate the maximum mass of potassium oxide that could be produced when 15.6 g of potassium reacts completely with oxygen. [3]

$$A_r(\text{K}) = 39$$

$$A_r(\text{O}) = 16$$

Mass = g

- (ii) Another reaction requires 0.050 mol of oxygen gas. Calculate the number of molecules in 0.050 mol of oxygen gas. Give your answer in **standard form**. [2]

$$\text{Avogadro's number} = 6.0 \times 10^{23}$$

Number of molecules =



10. (a) Tables **A**, **B**, **C** and **D** show the results recorded by four different groups in a class investigating the reactivity of Group 7 elements (the halogens).

Each group added iodine, bromine and chlorine to solutions of sodium halides.

A tick (✓) indicates when a reaction took place and a cross (×) indicates no reaction. Only **one** table shows the expected results for this experiment.

Table **A**

	sodium chloride	sodium bromide	sodium iodide
bromine	✓	✓	×
chlorine	✓	×	✓
iodine	✓	✓	✓

Table **B**

	sodium chloride	sodium bromide	sodium iodide
bromine	✓	✓	✓
chlorine	×	×	✓
iodine	×	✓	✓

Table **C**

	sodium chloride	sodium bromide	sodium iodide
bromine	×	×	✓
chlorine	×	✓	✓
iodine	×	×	×

Table **D**

	sodium chloride	sodium bromide	sodium iodide
bromine	×	✓	×
chlorine	✓	×	✓
iodine	×	✓	✓



- (i) Give the **letter** of the table which shows the expected results for this experiment. Explain why these are the expected results. [3]

Letter

.....

.....

.....

- (ii) Write a balanced symbol equation for the reaction between chlorine, Cl_2 , and sodium iodide solution. [3]

.....

- (b) A sample of iron bromide with a mass of 37.0 g contains 7.0 g of iron. Find the simplest formula of the iron bromide.

You **must** show your working. [3]

$$A_r(\text{Fe}) = 56$$

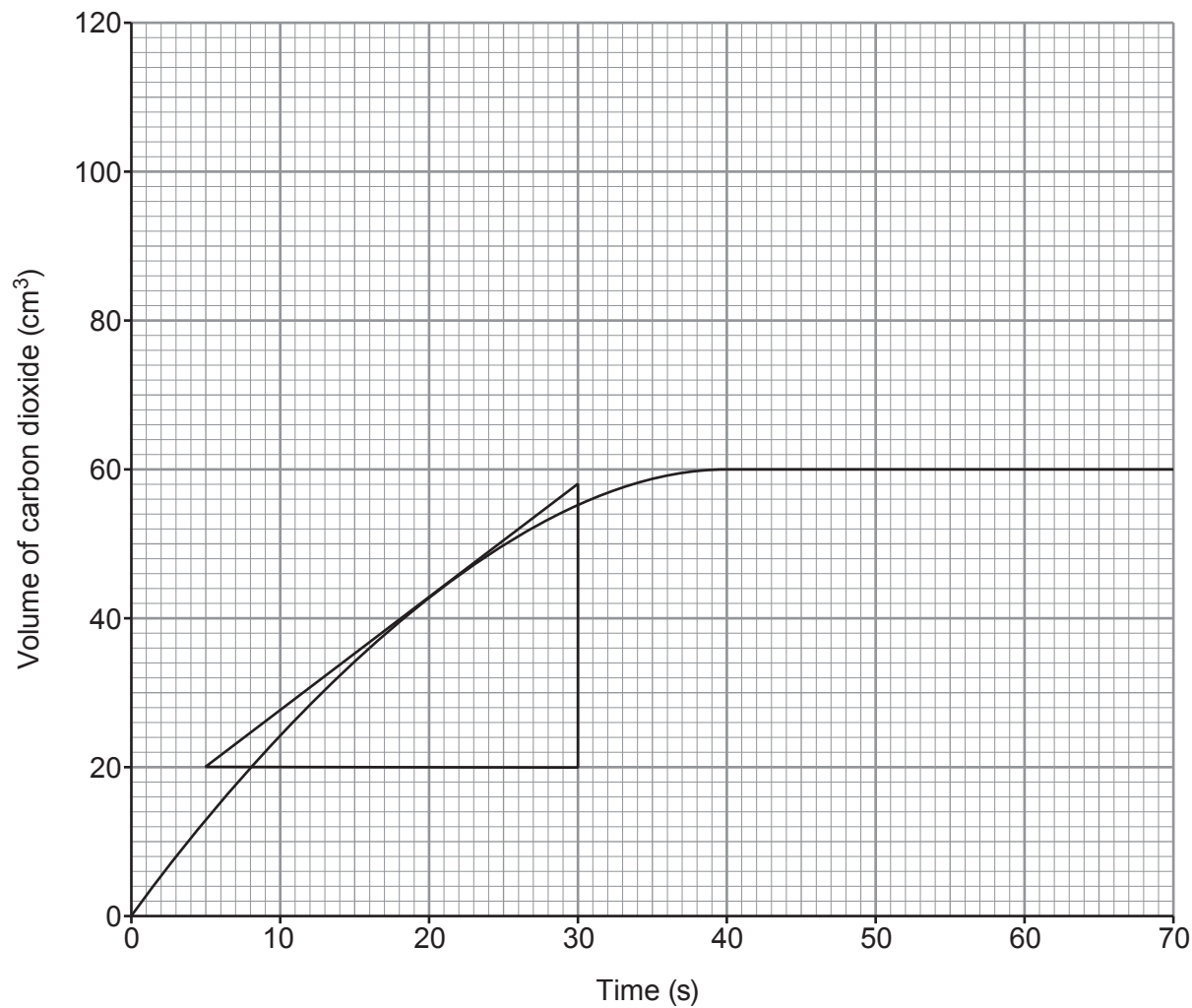
$$A_r(\text{Br}) = 80$$

Simplest formula

9



11. The graph shows the volume of gas produced in a reaction between 0.40 g of calcium carbonate powder and excess 1 mol/dm³ hydrochloric acid.



(a) Use the tangent drawn to calculate the rate of the reaction at 20 s. [2]

Rate = cm³/s



Examiner
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- (b) (i) On the same grid, sketch the graph that would be seen if the reaction were repeated with 0.60 g of calcium carbonate powder and acid of the same concentration and still in excess. [2]
- (ii) Explain the graph that you have sketched using your knowledge of particle theory. [2]

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END OF PAPER



[illegible]

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FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2

Group

3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2																									
7	Li	9	Be											11	B	12	C	14	N	16	O	19	F	20	Ne										
	Lithium		Beryllium												Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon										
	3		4												5		6		7		8		9		10										
23	Na	24	Mg											27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar										
	Sodium		Magnesium												Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon										
	11		12												13		14		15		16		17		18										
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
	19		20		21		22		23		24		25		26		27		28		29		30		31		32		33		34		35		36
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	103	Rh	106	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
	Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Palladium		Silver		Cadmium		Indium		Tin		Antimony		Tellurium		Iodine		Xenon
	37		38		39		40		41		42		43		44		45		46		47		48		49		50		51		52		53		54
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	192	Ir	195	Pt	197	Au	201	Hg	204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
	Caesium		Barium		Lanthanum		Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury		Thallium		Lead		Bismuth		Polonium		Astatine		Radon
	55		56		57		72		73		74		75		76		77		78		79		80		81		82		83		84		85		86
223	Fr	226	Ra	227	Ac																														
	Francium		Radium		Actinium																														
	87		88		89																														

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number